

Local and Global GANs with Semantic-Aware Upsampling for Image Generation (-Summary of Changes-)

We would like to thank the reviewers for providing constructive comments. We are pleased that they find our paper to be well-written and easy to follow and understand (**R2**, **R3**, **R4**). We are encouraged that they find our motivation and idea to be meaningful (**R2**, **R3**, **R4**), clear (**R2**, **R3**, **R4**), and novel (**R2**, **R3**, **R4**).

We have carefully addressed each comment below. The modified parts in the manuscript have been highlighted in blue.

Reviewer 1

To R1 Q1: *Title: “image generation” → “semantic/semantic-guided image generation” as they are different tasks.*

Yes, these two are different. This paper focuses on “semantic-guided/aware image generation”, but “semantic-aware” is already included in the title, so in order to avoid repetition, we simplified the title to “image generation”.

To R1 Q2: *Organization and Length: the application part (section 4.2.3) makes little sense to me as it does not show anything new or superior. The experiment part is too long and some of them can be moved into the supplementary materials.*

We have moved the content of section 4.2.3 to the appendices.

To R1 Q3: *Readability: the presentation of the manuscript is okay but far from great. A lot of effort is paid in describing the components of the proposed techniques but there is little discussion/reasoning on their motivations (e.g. section 3.2). In other words, there is a significant gap between very high-level intuitions and detailed implementations in the current presentation, which makes it difficult to understand. I suggest thorough rework the writing, especially section 3.2.*

Other reviewers **R2**, **R3**, and **R4** think our paper is well written and very easy to read, especially for section 3.2.

To R1 Q4: *Missing references: [1] Zhu, P., Abdal, R., Qin, Y. and Wonka, P., 2020. Sean: Image synthesis with semantic region-adaptive normalization. CVPR2020. [2] Zhu, Z., Xu, Z., You, A. and Bai, X., 2020. Semantically multi-modal image synthesis. CVPR2020. [3] Richardson, E., Alaluf, Y., Patashnik, O., Nitzan, Y., Azar, Y., Shapiro, S. and Cohen-Or, D. Encoding in style: a stylegan encoder for image-to-image translation. CVPR2021.*

We have cited these three works in our paper.

To R1 Q5: *No comparison with more advanced state-of-the-art, e.g. missing references [1] and [2]. These should be included in the evaluation.*

Both [1] and [2] use a different way to do evaluation. For instance, results of SPADE on Cityscapes are 62.3 mIoU, 81.9 accu, and 71.8 FID, but the paper [1] reports 57.88 mIoU, 93.59 accu and 50.38 FID, respectively. Results of SPADE paper on ADE20K are 38.5 mIoU and 79.9 acc, but the paper [2] reports 42.0 mIoU and 81.4 acc. For more details about [1], please refer to this github issue <https://github.com/ZPdesu/SEAN/issues/8>. For more details about [2], please refer to this github issue <https://github.com/Seanseattle/SMIS/issues/6>.

Thus, for a fair comparison, we adopt the same training/test dataset split, model hyper-parameters (e.g., the number of training epoches), and evaluation method of ours. We have added the results of [2] (i.e., SMIS) in Table 5 and Figure 11(a) and Figure 11(b). We have added the results of [1] (i.e., SEAN) in Table 5 and Figure 11(c).

To R1 Q6: *The motivation of SAU is not convincing. The environment can also have an impact on object appearance (e.g. lighting and shadow). So why exclude them from the strategy?*

This is a common problem of semantic image synthesis methods. The segmentation map can only provide the shape and structure information of the object, and cannot provide both lighting and shadow information.

To R1 Q7: *How are the filtered feature maps in Fig.6 visualized? What do the colors mean?*

As indicated in page 11, we randomly plot some channels from the learned class-specific feature maps (30th to 32th, and 50th to 52th) on Cityscapes to see if they clearly highlight particular semantic classes. Different colors mean different classes such as road, vegetation, and cars. We can easily observe that each local subgenerator effectively learns the deep class-level representations, further confirming our motivations.

To R1 Q8: *Section 3.2.1, How is the “feature shuffle” performed exactly? How are the $k^2 s^2 \times H \times W$ rearranged to $k^2 \times Hs \times Ws$? Along which dimension?*

Feature shuffle is a variant of pixel shuffle, performed on feature maps. Pixel shuffle layer is one of the most recent layer types introduced in modern deep learning Neural Networks. Its application is closely related to the single-image super-resolution (SISR) research, i.e., the technique ensemble whose aim is to produce a high-resolution image from a single low-resolution one. For more details about the pixel shuffle layer, please refer to Figure 1 of the reference [1].

To R1 Q9: *If SAU aims to use information from the same class, why not take the semantic mask as an input? If it is learned, why it favors pixels from the same class?*

The task of semantic image synthesis is to input a semantic mask, and then output a photo-realistic image. So our model uses a semantic mask as input. However, the existing upsampling methods such as deconvolution, pixel shuffle, and spatial attention cannot well maintain the semantic information between the same class, so we propose the SAU to alleviate this problem.

To R1 Q10: *For SAU, if there is still a kernel of size k , how can it represent long-range pixels in same class?*

This kernel of size k is to balance performance and efficiency. We can set k to be larger, which can capture pixels with longer distances between the same class, but at the same time increase the training time.

To R1 Q11: *I do not understand Fig. 5.*

Please refer to section 3.2.3 for more details. Moreover, **R2** thinks Figure 5 is highly interesting.

To R1 Q12: *Fig. 12, how are the feature maps visualized? What do the colors mean?*

See **R1 Q7** for more details.

Reviewer 2

To R2 Q1: *According to the contributions listed on Page 3-L40-Left, the authors also mentioned a more detailed related work. However, two works that highly overlap with LGGAN such as MaskGAN [45] and SEAN [-1] (both presented in the same conference as LGGAN, CVPR’20), were not fully addressed nor even included in the quantitative or qualitative evaluation. In detail, MaskGAN is only mentioned on Page 3-L39-Right. The authors argue that MaskGAN has only been applied for*

faces, and LSGAN is a more general-purpose technique. Although MaskGAN has only been tested on faces, this is highly misleading because similar to LSGAN, MaskGAN uses semantic information to generate RGB images, and it is not limited to faces. Therefore, a direct comparison would have been expected with MaskGAN and SEAN.

We conducted MaskGAN and SEAN experiments on both Cityscapes and ADE20K datasets. We have included all these results of both MaskGAN and SEAN in Tables 5, 6, 7, and 8, and Figures 11 (c) and Figures 11 (d).

To R2 Q2: Furthermore, Page 3-L56-Left states that extensive ablation studies demonstrate the effectiveness of the proposed SAU. However, only FID (why not mIoU and Acc as well?) quantitative comparison is shown in Table 10 with respect to different upsampling techniques. In my opinion, an ablation study aims at breaking down the proposed technique and highlight the important parts of it, and this is not the case here.

We have included the results of both mIoU and Acc in Table 12. The proposed SAU cannot be broken down, it cannot work without one of the parts. Therefore, we cannot do more ablation studies about the proposed SAU. Moreover, we have provided a very detailed ablation study about our framework in Table 9.

To R2 Q3: Finally, the proposed SAU is coupled with GauGAN and LSGAN, and a quantitative evaluation is shown in Table 6. Although it consistently outperforms GauGAN, it is not the case for LGGAN. mIoU and Acc are worse for both Cityscapes and ADE20K. Why is that? It is a bit surprising because one of the motivations of SAU is to better preserve semantic information. Results with MaskGAN+SAU and SEAN+SAU would be very interesting here.

Because our LGGAN framework can achieve very high mIoU and Acc results, adding SAU may slightly reduce the result of mIoU and Acc, but greatly improve the result of FID. Moreover, after adding the proposed SAU to the original SEAN, MaskGAN, and GauGAN, results of all three evaluation metrics are further improved.

We have provided the results of both MaskGAN+SAU and SEAN+SAU in Table 5, Figure 11(c) and Figure 11(d).

To R2 Q4: As a minor issue, it would be interesting to also see the runtime with SAU.

We test the proposed SAU on an NVIDIA V100 GPU and achieve 18.9 fps on Cityscapes.

Reviewer 3

To R3 Q1: This paper extends an earlier work that was published in CVPR 2020. I believe that the current version has enough additional contribution to be accepted to TPAMI. The technical contributions are strong, and the additional upsampling module is very interesting. It would be interesting to see if the upsampling module is applicable to other tasks, e.g. super-resolution, segmentation, etc. The writing is excellent and the authors provide in-depth evaluation and analysis.

Because of the limitation of the number of pages, applying our upsampling module to other tasks such as super-resolution and segmentation is our future work.

To R3 Q2: One minor thing I would like to point out is the user study. I am not sure whether the user studies strongly suggest a clear preference with numbers like 54%, when the choice is binary. What happens when the user studies are conducted with choosing the best one out of many choices? SelectionGAN seems to show good quantitative numbers, so it would also be interesting to include SelectionGAN in the user study. I recommend a minor revision with the emphasis of more extensive user study.

We have included the results of SelectionGAN in Table 6. We can see that our LGGAN achieves better results than SelectionGAN in Table 6.

Moreover, we have added more user study results in Table 7. We see that the proposed LGGAN++ achieves significantly better results than existing methods.

Finally, according to your request, we have added additional user study results in Table 8. Participants are asked to pick the best realistic image among many choices, and we give them unlimited time to respond. Each trial consists of 30 comparisons and is performed by 30 distinct participants. We observe that more participants consider our results to be more realistic than those provided by the other methods.

We have included these results and discussions in page 10-11.

Reviewer 4

To R4 Q1: *The experiment results do not convince me that the presented approach significantly outperforms other baselines. For example, in Table 6, the user study only suggests that the results by LGGAN are comparable to Pix2pixHD, SIMS, and GauGAN.*

We have added more user study results in Table 7. We see that the proposed LGGAN++ achieves significantly better results than existing methods.

Moreover, according to the request of **R3**, we have added additional user study results in Table 8. Participants were asked to pick the best realistic image among many choices, and we give them unlimited time to respond. Each trial consists of 30 comparisons and is performed by 30 distinct participants. We observe that more participants consider our results to be more realistic than those provided by the other methods. For more details, please refer to **R3 Q2**.

To R4 Q2: *I am very confused that LGGAN is used for comparison to other baselines instead of using LGGAN++. It seems to me that LGGAN++ is expected to be better than LGGAN, but the authors present the results of LGGAN in several figures in Fig. 1,8,9,10. Also, for the application results in Fig. 19, it is not clear if the result is obtained by LGGAN or LGGAN++. In Table 6, this major quantitative evaluation is also on LGGAN.*

Figures 1, 8, 9, and 10 are the results of cross-view image translation. We find that LGGAN has achieved state-of-the-art results on the cross-view image translation task, so we use LGGAN to compare with other existing methods in this task.

The results of Figure 19 are obtained by using LGGAN++. We have added more results in Tables 7 and 8 to validate the effectiveness of the proposed LGGAN++.

To R4 Q3: *I am not very convinced that SAU can lead to better results for LGGAN. Looking at the results of Table 5, I think the difference is minor after adding SAU.*

Because our LGGAN framework can achieve very high mIoU and Acc results, adding SAU may slightly reduce the result of mIoU and Acc, but greatly improve the result of FID. Moreover, after adding the proposed SAU to the original SEAN, MaskGAN, and GauGAN in Table 5, results of all three evaluation metrics are further improved, which further validates the effectiveness of the proposed SAU.

To R4 Q4: *The idea of SAU is very similar to the involution operation proposed in this paper (except for adding the upsampling design): Involution: Inverting the Inherence of Convolution for Visual Recognition (arXiv:2103.06255, CVPR 2021) What if we simply apply “involution+ bilinear upsampling” or “involution+ bilinear upsampling”? Do we achieve the same results?*

We have added the results of “involution+bilinear upsampling” in Table 12 and Figure 18. We can see that our SAU achieves better results than the involution method.

References

- [1] W. Shi, J. Caballero, F. Huszár, J. Totz, A. P. Aitken, R. Bishop, D. Rueckert, and Z. Wang, “Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network,” in *CVPR*, 2016. 2